

PessAssist Device (PAD)

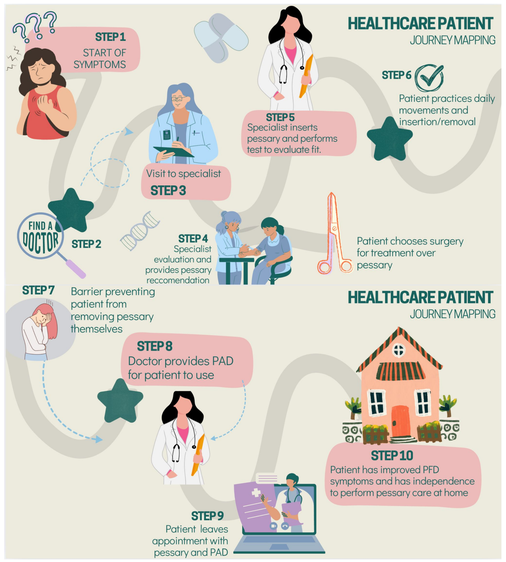
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Biomedical Engineering

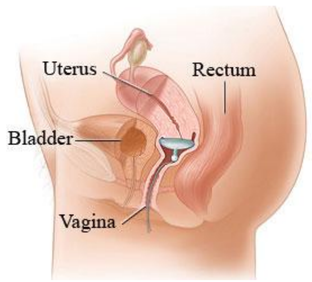
Background & Motivation

- Approximately 50% of women will experience pelvic organ prolapse (POP) [1]
  - Over 51% of patients prefer non-surgical treatments to avoid high costs and risks [2]
- The Gellhorn pessary is a common option, but its unique geometry makes it very difficult to remove, especially for those with limited dexterity, strength, or reach
  - Tools to aid in removal exist but are targeted toward ring pessaries



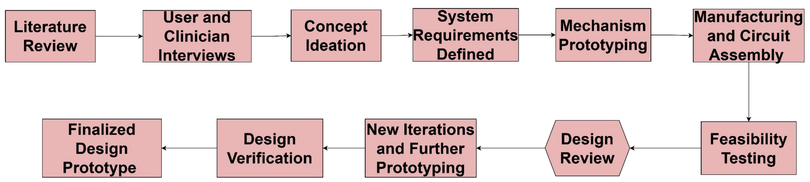


Gellhorn Pessary

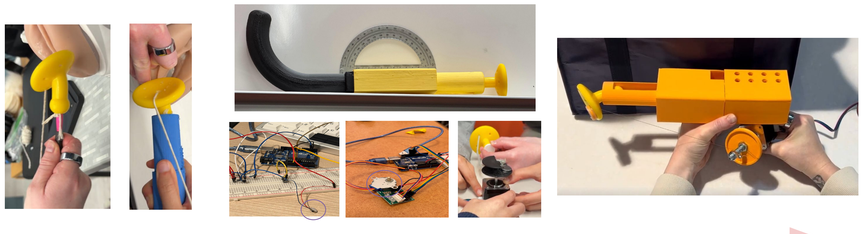


MedGyn Ring Pessary Remover

Iterations

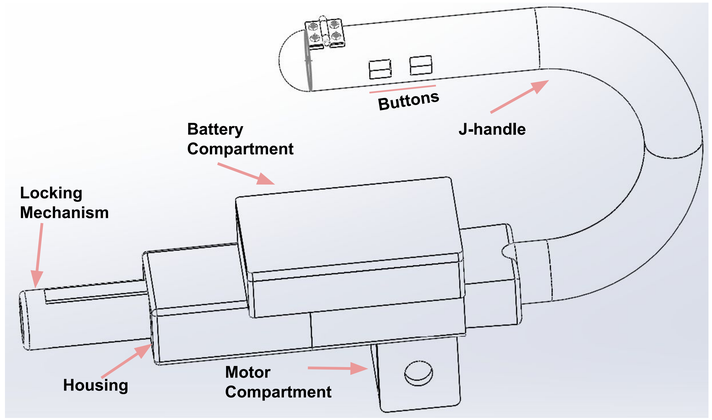


Evolution of PAD



Design Solution

PAD Device

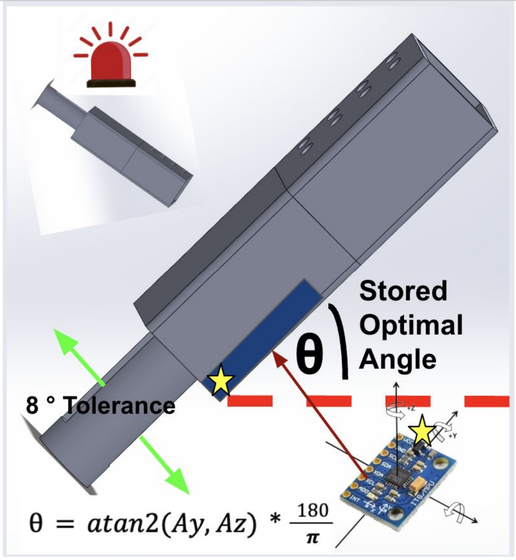


Handheld Probe

- Houses the internal electrical components
- J-shaped handle to allow user to maneuver PAD while in semi-Fowler position

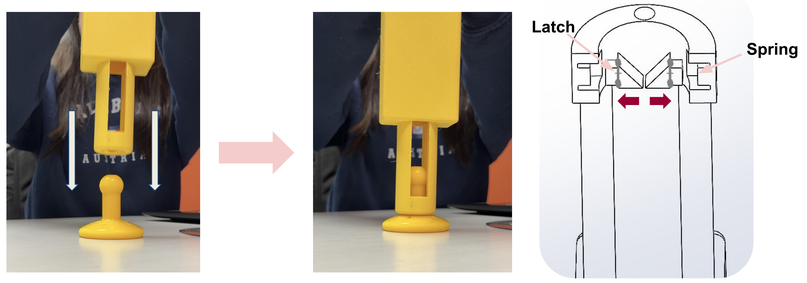


- The roll angle, the rotation about the x-axis, is measured by the MPU6050 and is compared to the patient's optimal angle for insertion. An alarm is delivered if the angle is beyond the safety threshold



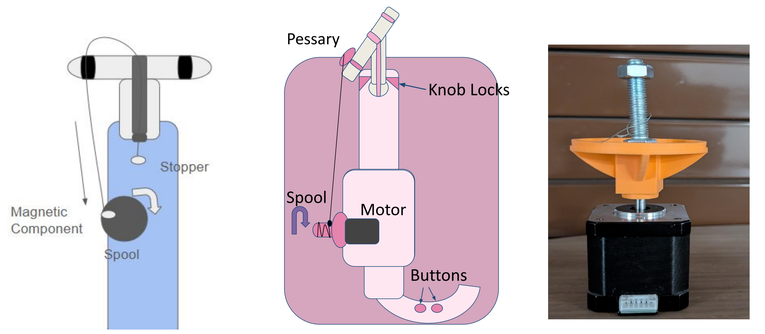
Gellhorn Locking Mechanism

- Secures the probe onto the stem of the pessary with spring-latch mechanism
- Ensures probe remains in stable position during automated suction removal



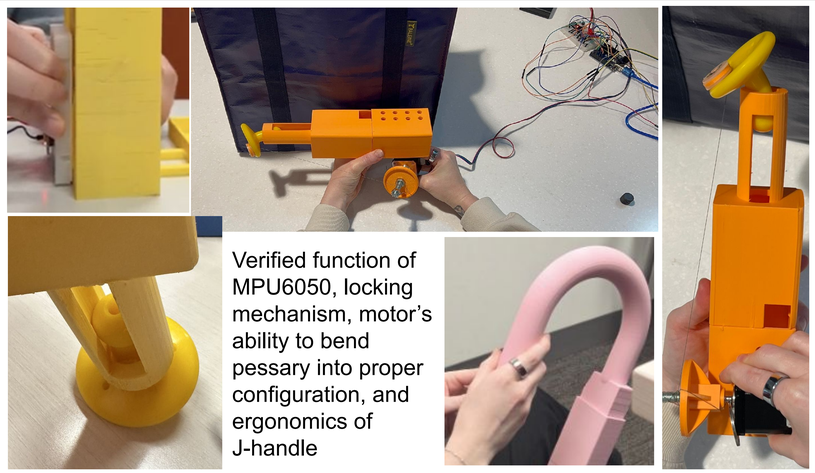
Suction Remover

- Automated suction-release mechanism to break seal of pessary from prolapsed tissue
- Assists in bending the pessary to the appropriate configuration for easier and safer removal



Design Verification Results

Design Verification



Verified function of MPU6050, locking mechanism, motor's ability to bend pessary into proper configuration, and ergonomics of J-handle

Conclusions & Future Directions

- Pessary removal devices will allow a safe, self-guided option for patients that were previously clinician-dependent for pelvic health.
- A future iteration of PAD would introduce a clinician-guided orientation app. This would allow clinicians to define optimal insertion/removal angles with real-time feedback.
- Automated, motor-driven-suction removal with an optimized run time to properly collapse the pessary would reduce the need for user-dependence.

References

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1. "Demystifying Pelvic Organ Prolapse." www.uchicagomedicine.org, www.uchicagomedicine.org/forefront/womens-health-articles/demystifying-pelvic-organ-prolapses.
2. Bodner-Adler, Barbara, et al. "Prolapse Surgery versus Vaginal Pessary in Women with Symptomatic Pelvic Organ Prolapse: Which Factors Influence the Choice of Treatment?" Archives of Gynecology and Obstetrics, vol. 299, no. 3, 17 Jan. 2019, pp. 773–777, https://doi.org/10.1007/s00404-019-05046-7.